Standard Errors and Clustering: One Estimate, Four Intervals

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 4, Hour 1 (slides 126 to 129). Same story as Day 3: a tutoring programme at one school.

Step 1. The story

Tutoring was given to **whole classes**, not to individual students. Four classes of five. Classes 1 and 2 got tutoring; classes 3 and 4 did not. The estimate is the same whatever we do next. The standard error is not.

Class	Tutoring	Scores	Class mean
1	yes	78, 80, 82, 84, 86	82
2	yes	68, 70, 72, 74, 76	72
3	no	68, 70, 72, 74, 76	72
4	no	62, 64, 66, 68, 70	66

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τ̂= mean(tutored) − mean(untutored) = 77 − 69 = +8
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Step 2. Four standard errors for one estimate

Neyman (Day 2) treats the 20 students as 20 independent coin flips:

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SE = √(S₁²/n₁ + S₀²/n₀) = √(36.7/10 + 18.9/10) = 2.36 CI: 8 ± 1.96 × 2.36 = [3.4, 12.6]
```

But there were not 20 coin flips. There were two: which two classes got tutoring. Students in the same class share a teacher, a room, a timetable, so their scores move together. Clustered SE treats the **class** as the unit. Four class means, two per arm:

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tutored means 82, 72 (variance 50) untutored means 72, 66 (variance 18)
SE = √(50/2 + 18/2) = 5.83 CI: 8 ± 1.96 × 5.83 = [−3.4, 19.4]
```

Standard error	Treats as independent	SE	95 percent CI	Contains zero?
Classical (OLS default)	20 students, equal variance	2.36	[3.4, 12.6]	no
Robust HC2	20 students, unequal variance	2.36	[3.4, 12.6]	no

Clustered by class	**4 classes**	**5.83 (regression cluster-robust: 4.89)**	**[−3.4, 19.4]**	**yes**
Bootstrap, resample students	20 students	2.35	[3.4, 12.6]	no
Bootstrap, resample classes	**4 classes**	**4.5**	**[−0.8, 16.8]**	**yes**

Same +8. With the wrong SE the result is significant; with the right one it is not. The first two rows count correlated students as if each brought new information. Slide 127 calls this the most common uncertainty error in applied work.

Step 3. The rule

> `Cluster at the level where treatment was assigned. Never finer.`

Assigned by	Cluster by	Example
student (coin per student)	nothing extra; Neyman or HC2 is right	Day 2 energy drink
class	class	this handout
school	school	the DiD library study
region and month	region	Lab 3 regional policy

Four clusters is too few for the usual formula to be trusted (slide 127): the honest report is the estimate, the clustered interval, and the sentence "only four clusters, so the interval is approximate". Lab 4 uses a wild cluster bootstrap for this case.

Step 4. Reading the interval (slide 129)

[−3.4, 19.4] does not say "tutoring does nothing". It says the data are compatible with anything from a small loss to a large gain. Wide interval means underpowered, and the fix is more classes, not more students inside the same classes: 40 students in the same four classes leaves the SE almost unchanged.

Three things to remember

1. The estimate does not depend on the standard error; the conclusion does.

2. Independence is counted in units of assignment, not rows of data.

3. Bootstrap the same unit you would cluster on.